

SHIVAM KUMAR

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SUMMARY

Software Development Engineer II at Microsoft (Azure Container Apps) with 5+ years building cloud-native platforms, AI-powered SRE agents, and CNCF open-source infrastructure (Dapr). Shipped LLM-based RCA agents that cut P50 incident time-to-mitigate by **97%** (185→6 min) and time-to-resolve by **64%**; led multi-region rollouts touching **~400 AKS clusters** and **~300 external customers**; architected internal forks and OneBranch/Ev2 pipelines for 5 Dapr repositories with **<1 day OSS release SLA**.

Focus: Distributed Systems · AI/LLM Agents · Site Reliability (SRE) · CI/CD & Release Engineering · Kubernetes & Container Platforms

EXPERIENCE

Software Development Engineer II

Jul 2024 – Present

Microsoft – Azure Container Apps

Hyderabad, India

- Designed and deployed **AI-powered SRE agents** for Azure Container Apps incident diagnostics, reducing P50 time-to-mitigate by **97% (185→6 min)** and time-to-resolve by **64%**; evolving toward **autonomous auto-repair** with automated RCA comparison; enabled **Foundry Control Plane** for ACA-hosted agents.
- Built **Session Management APIs** and fixed critical async execution bugs for ACA Sessions, improving stability for **~300 external customers**; enabled early session deletion for the Fabric team, optimizing billing and costs.
- Expanded **Azure Functions on ACA** platform — function keys management, invocation tracing, durable-events logging, KEDA scaling rules, and drain functionality, benefiting **~1,322 external apps**.
- Led **Dapr AKS Extension** rollouts (v1.15, v1.16) across all Azure regions; remediated CVEs across **~400 AKS clusters**; integrated CodeQL for pipeline security.
- Integrated **Azure Key Vault** with Managed Environment Storage API, securing **~27K daily API calls** by eliminating raw connection strings.
- Served as **on-call DRI** resolving **200+ incidents** including 27+ CRIs, outperforming team baselines.

Software Development Engineer

Jul 2021 – Jul 2024

Microsoft – Azure Container Apps / Dapr

Hyderabad, India

- Contributed to **Dapr** (CNCF open-source project) — improved several components and APIs to stable; implemented **cascade terminate/purge** for Dapr Workflows across runtime and SDKs; added bulk subscribe support in Azure Event Hubs; added workflow performance tests identifying P0 bugs.
- Enabled **airgap installation**, winget, and MSI for Dapr CLI; published combined coverage reports; resolved customer issues on GitHub and Discord.
- Architected **internal fork and CI/CD infrastructure** across 5 Dapr repositories; automated OSS-to-internal releases within **<1 day SLA**; consolidated into **OneBranch-compliant** pipelines with Ev2 deployment.
- Added **multi-arch support** (ARM64, ARM, AMD64), Mariner and distroless images; handled 1P Container Registry migration; delivered nightly builds for ACA; contributed to AKS Extension GA requirements.
- Helped launch **ACA Sessions public preview** — added support for Node.js code interpreter sessions; fixed critical bugs in Python and Node.js sessions; added performance tests identifying regressions.

Quantitative Research Intern

May 2020 – Jul 2020

JP Morgan & Chase – E-Trading

Mumbai, India (Remote)

- Built price prediction models using **LSTM**, feedforward neural networks, and regression with univariate feature selection.
- Improved model predictive power by **40%** for HK names through feature standardization and data preprocessing.
- Implemented L1 and L2 features for optimizing model parameterization across multiple asset classes.

TECHNICAL SKILLS

Languages: Go, Python, C#/.NET, JavaScript/TypeScript, C/C++

Cloud & Infrastructure: Azure (Container Apps, AKS, Functions, Key Vault, Event Hubs, CosmosDB), Kubernetes, Docker, Dapr (CNCF), Helm

Tools & CI/CD: OneBranch (1ES), Ev2, Azure DevOps Pipelines, GitHub Actions, Git, Geneva, CodeQL, KEDA

Domains: Distributed Systems, AI/LLM Agents, CI/CD & Release Engineering, Microservices, Site Reliability (SRE)

PROJECTS

Java Compiler | Python, PLY, Compiler Design (CS335)

Jan 2020 – Apr 2020

- Designed lexer, parser, and 3-address code generation; support for symbol tables, classes, functions, and interfaces.

GemOS | C, x86, Operating Systems (CS330)

Aug 2019 – Nov 2019

- Implemented mmap, munmap, mprotect with lazy allocation and pagefault handling; cfork/vfork with copy-on-write.

Machine Learning | Python, Keras, ML (CS771)

Aug 2019 – Nov 2019

- CNN image classification using Keras; SGD for binary classification; multi-label Bonsai recommender system.

MERN Mobile App | React Native, Node.js, MongoDB (CS252)

May 2019 – Jul 2019

- Full-stack mobile app with React Native frontend, Node.js/Express backend, MongoDB, and secure JWT login.

RESEARCH

Data Race Detection for Task-Parallel Programs

Jun 2019 – Aug 2021

Supervisor: Prof. Swarnendu Biswas, IIT Kanpur

- Designed novel algorithm **Tasker** integrating vector clocks with tree-based techniques; implemented using **LLVM** pass for memory instrumentation on task-parallel programs.
- Achieved **1.48X speedup** over PTRacer baseline on 128GB Intel Xeon system with lower memory overhead.
- “Efficient Data Race Detection of Structured Task Parallel Programs Using Vector Clocks” — S. Kumar, A. Agrawal, S. Biswas | Submitted to ISCGO 2022.

EDUCATION

Indian Institute of Technology, Kanpur

CGPA: 9.4/10.0

Bachelor of Technology in Computer Science and Engineering

2017 – 2021

ACHIEVEMENTS & LEADERSHIP

RockStar Award, Microsoft ('22, '23) for exceptional contributions | **Academic Excellence Award** (×2), IIT Kanpur

JEE Advanced **AIR 715** | JEE Mains **AIR 348** | KVPY Scholar | NSE Physics **Top 1%** | Ram Prakash Chopra Scholarship

Academic Mentor, IIT Kanpur ('18-19) | Secretary, Dramatics Club, IIT Kanpur ('18-19)